

# Middleware Lab: Queue vs Topic Messaging Models

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Duration : 30 Min.

## Scenario

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A messaging broker design must handle point-to-point transactional order updates and broadcast notifications simultaneously.

## Task Objectives

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Perform the following actions inside the `student_workspace` directory: - Draft a comparison guide detailing queue (point-to-point) and topic (publish-subscribe) communication models.

## Instructions to Perform the Task

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1. When your workspace loads in **VS Code**, use the **Explorer** panel on the left to locate your files.
2. Navigate to and click the specific files mentioned in the Task Objectives.
3. Make the necessary code edits in the editor.
4. Press `Ctrl + S` (Windows) or `Cmd + S` (Mac) to save your changes.
5. If you need to run commands, open the built-in terminal by clicking **Terminal > New Terminal** from the top menu.

## Validation

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Once you have saved your files and are ready to submit, return to the platform dashboard and click the **"Submit Final"** button. This task is evaluated manually by the grading team.

## Evaluation Rubric:

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1. Defining characteristics of Point-to-Point Queue model (FIFO, single consumer): 3 marks
2. Defining characteristics of Publish-Subscribe Topic model (fan-out, multiple consumers): 3 marks
3. Selecting appropriate model for e-commerce checkouts and log broadcast feeds: 4 marks

**Total Score: 10 Marks**

## Important Notes

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- This is a manually evaluated question.
- Ensure all logs and configurations are fully saved in your workspace folder before submitting.